

MANAGERIAL ECONOMICS

Production Observation & Bakery Visit Report

Case Study: Bake Studio

Name: Palak Bansal

Programme: PGDM – Financial Management

Roll No: 26PGDMFM024

Date of Visit: 30 August 2026

Interviewee: Mr. Shivam Arora (Co-Founder at BAKE STUDIO)

1. Business Overview

Business Name	Bake Studio
Established	2020
Nature of Business	QSR (Quick Serving Restaurant) + Bakery
Team Size	Approximately 150
Permanent Employees	Approximately 130–135
Temporary Workers	No regular temporary workforce reported
Normal Daily Output	Approximately ₹5–7 lakh per day
Peak / Festival Output	Approximately ₹20–25 lakh per day
Interviewee	Mr. Shivam Arora

Bake Studio combines bakery production with a quick-serving restaurant format and focuses on products suited to current consumer preferences. The visit was used to understand how labour, capital, technology and demand interact in the production process.

2. Interview with Personnel – Question & Answer

Q1. How many years has the bakery been operating, and what are its main products?

Bake Studio has been operating since 2020. It operates as a QSR along with a bakery and focuses on products preferred by Gen Z customers.

Q2. Approximately how much do you produce on a normal day and during peak demand periods?

Normal production is approximately ₹5–7 lakh per day. During festivals and high-demand periods, it can increase to around ₹20–25 lakh per day.

Q3. How many permanent and temporary workers do you have, and what are their main roles?

There are approximately 130–135 permanent employees and no regular temporary workforce. Employees are mainly divided across the cake department (20–25), sales department (50–55) and factory operations (around 60).

Q4. What skills are required, and what challenges do you face in hiring workers?

Employees need basic reading skills to understand recipes, along with adaptability, willingness to work, dedication, hard work and leadership. Finding people with the required basic skills, willingness and adaptability is a challenge.

Q5. If you hire more workers, does production increase proportionately? Why or why not?

Production may increase, but adding more workers can make management chaotic and may not achieve the expected output. The business therefore prefers skilled workers rather than simply increasing headcount.

Q6. What major machines and equipment do you use, and which are most important for production?

Major equipment includes ovens, bread slicers, planetary mixers, proofers, dough sheeters and an automated cookie dropper. All are important to the production system.

Q7. Which parts of production are manual, semi-automated, or automated?

Most products follow a semi-manual process in which machines and people work together.

Q8. Do you use any digital tools or software for billing, inventory, or online orders?

Yes. The business uses Petpooja, CRM software and Zomato.

Q9. Has machinery or technology improved production speed, quality, or reduced wastage/labour requirements?

The business uses data-based work, with demand and production maintained through data analysis. According to the interview, this has improved overall operations.

Q10. Can you briefly explain the production process from raw materials to the finished product?

For cake production, raw materials are procured, premixes and ingredients such as butter, oil and water are mixed using a planetary mixer, the mixture is moulded and baked, and then the sponge is combined with cream and toppings, assembled and given final finishing. Production is planned according to outlet-wise demand.

Q11. What is the biggest limitation or bottleneck affecting your production capacity?

The biggest limitation identified was a lack of skilled workers.

Q12. What do you do to improve productivity and efficiency?

The business focuses on high-end quality, hygiene, store hygiene, efficient use of raw materials and store productivity to attract customers.

Q13. If you increase workers, machines and other inputs, does output increase more than proportionately, proportionately, or less than proportionately?

The manager explained that there must be sufficient demand. When demand increases, production also increases gradually. Therefore, the scale of production depends strongly on demand.

Q14. Have you experienced any benefits or problems when increasing production on a larger scale?

Higher volume can provide benefits such as better raw-material pricing and better rates for customers. Increasing demand and production also supports consistency.

Q15. What is the one biggest improvement you think the bakery needs to increase efficiency or production?

The interview highlighted the importance of authentic raw materials and genuine products, particularly considering customers' ability to pay in Tier-2 and Tier-3 cities.

3. Production Process – Cake Production

Based on the interview, the cake production process can be represented as follows:

1	2	3	4	5	6	7	8
Raw Material Procurement	**Premixes & Ingredients**	**Planetary Mixing**	**Moulding**	**Oven Baking**	**Sponge + Cream + Toppings**	**Assembly & Finishing**	**Outlet-wise Dispatch / Sale**

Key process observation: The process is largely semi-manual, with employees and machines working together. Production is also adjusted according to the demand of individual outlets.

4. Production Inputs: Labour, Capital & Technology

Factor	Observation	Economic Meaning	Impact
Labour	130–135 permanent employees; skilled and adaptable workers are preferred.	Labour is a key variable input.	Skill level affects productivity and coordination.
Capital	Ovens, mixers, proofers, dough sheeters, slicers and automated cookie dropper.	Capital supports production capacity.	Machinery enables higher-volume and consistent production.
Technology	Semi-manual production; Petpooja, CRM and Zomato; data-based demand planning.	Technology improves coordination and information use.	Supports efficiency and demand-based production planning.

5. Link with Managerial Economics

5.1 Production Function

The bakery's output can be understood through the Production Function, where output depends on the combination of inputs such as labour, capital and technology. In this case, workers, bakery machinery and digital systems jointly contribute to production.

Cobb–Douglas form: $Q = A L^{\alpha} K^{\beta}$

Here, Q represents output, L represents labour, K represents capital, and A represents the contribution of technology/productive efficiency. The visit provides qualitative evidence of these inputs, but not the numerical data required to estimate α and β.

5.2 Law of Variable Proportions

The interview suggests that adding more workers does not necessarily increase output proportionately. Beyond an effective level, additional workers can create coordination and management difficulties. The bakery therefore emphasises skilled labour rather than simply increasing the number of employees.

5.3 Productivity

Productivity is improved through skilled employees, machinery, hygiene, quality control, efficient raw-material use and data-based demand planning. The key insight is that productivity is about the efficient use of resources, not only about increasing the number of inputs.

5.4 Returns to Scale

Based on the interview, Bake Studio shows indications of increasing returns to scale when demand supports expansion: larger volumes can create raw-material pricing benefits, better rates for customers and greater

consistency. However, the manager clearly stated that scaling is demand-dependent, so expansion without sufficient demand would not be justified.

6. Key Observations & Learning

- The bakery operates with a large permanent workforce, but skilled labour is more valuable than simply adding more workers.
- Production is semi-manual, showing a complementary relationship between labour and machinery.
- Technology supports demand forecasting/planning and operational efficiency.
- Output varies substantially with demand, rising from about ₹5–7 lakh normally to ₹20–25 lakh during festivals.
- The main production bottleneck identified is the shortage of skilled workers.
- During very high demand, operational fumbles can occur, showing the importance of coordination and capacity planning.

7. Evidence-Based Recommendations

1. Strengthen skilled-worker training

Since lack of skilled workers was identified as the biggest production limitation, structured training and skill development can reduce dependency on a limited pool of skilled employees.

2. Use demand-based workforce and capacity planning

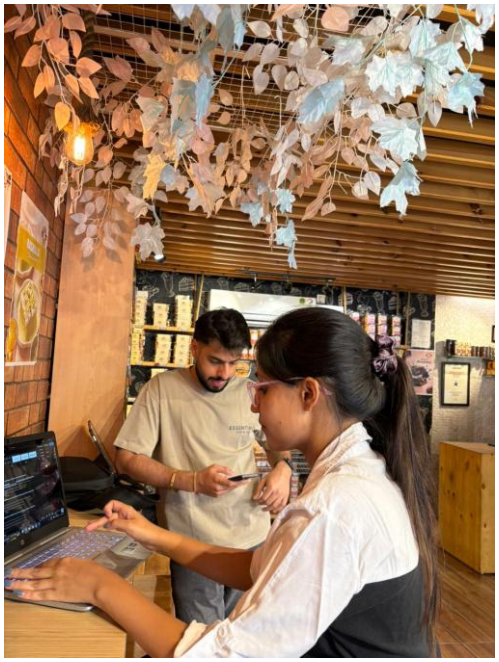
Since production is strongly linked to demand and high-demand periods can create operational fumbles, staffing and production schedules should be planned in advance using the existing data-based approach.

3. Maintain raw-material quality while managing cost

The interview highlighted the importance of authentic raw materials and genuine products. The bakery can use supplier evaluation and demand-based procurement to maintain quality while controlling input costs.

8. Photographic Evidence of Bakery Visit

The following photographs document the bakery premises, production activity and the interview/interaction during the visit.



Interview / interaction at the bakery



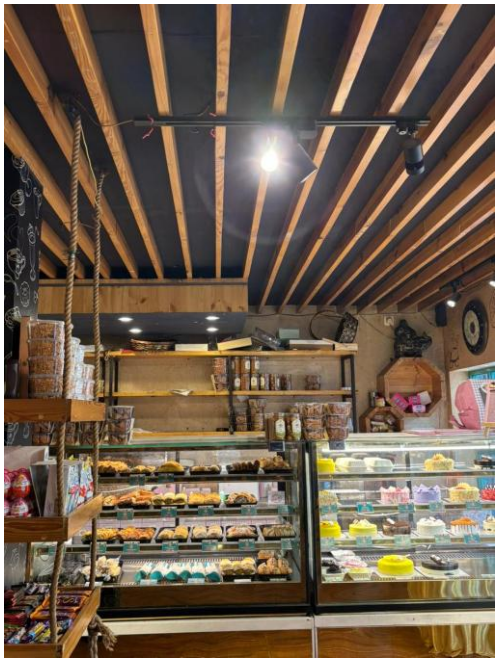
Interview with bakery personnel



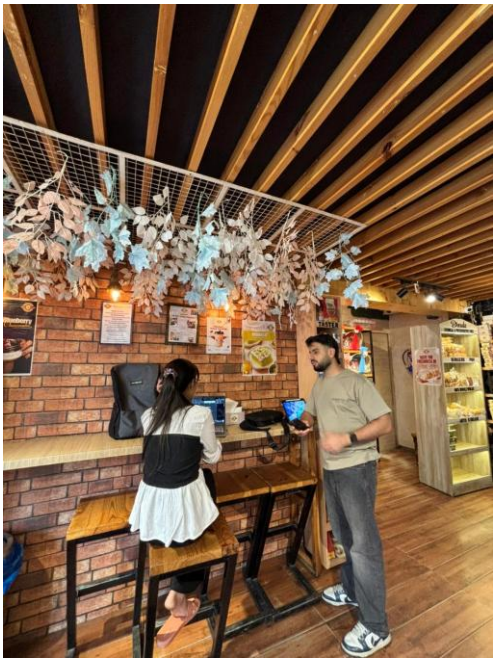
Observation of cake-production activity



Observation of production area



Bakery outlet and display area



Interaction during the visit



Bakery premises and customer-facing area



Business recognition / customer-related display

9. Three-Minute Explainer Video

The video explains the bakery visit, key observations, understanding of the production process, interview highlights, and the connection between the observations and Managerial Economics concepts.

Google Drive Video Link:

https://drive.google.com/file/d/1pZXKLmdZh01JptpKKv-zYRsTgqz8mjpV/view?usp=sharing

10. Conclusion

The visit to Bake Studio helped connect classroom concepts with an actual production environment. The bakery demonstrates how labour, capital and technology work together, while demand determines the scale at which production should operate. The most important managerial insight from the visit is that increasing inputs alone does not guarantee higher productivity; the right mix of skilled labour, machinery, technology and demand-based planning is essential.